

DSA MySQL CAPSTONE PROJECT: OLIST BRAZILLIAN E-COMMERCE DATASET

1. Who are the top 10 customers by total amount spent?

```
-- 1. top 10 customers by total amount spent
use olist;

select 3 s
    c.customer_unique_id,
    format(sum(op.payment_value), 2, 'en_US') as total_spent
from customers c
join orders o
    on c.customer_id = o.customer_id
join order_payments op
    on o.order_id = op.order_id
group by c.customer_unique_id
order by sum(op.payment_value) desc
limit 10;
```

2. What is the monthly revenue trend across the dataset?

```
-- 2. monthly revenue trend
use olist;

select
    date_format(o.order_purchase_timestamp, '%Y-%m') as order_month,
    format(sum(op.payment_value), 2) as monthly_revenue
from orders o
join order_payments op
    on o.order_id = op.order_id
group by date_format(o.order_purchase_timestamp, '%Y-%m')
order by order_month;
```

3. What is the month-over-month change in revenue?

```
-- 3. month over month change in revenue
with monthly_revenue as (
  select
    date_format(o.order_purchase_timestamp, '%Y-%m') as order_month,
    sum(op.payment_value) as revenue
  from orders o
  join order_payments op
  on o.order_id = op.order_id
  group by date_format(o.order_purchase_timestamp, '%Y-%m')
)
select
  order_month,
  format(revenue, 2, 'en_US') as monthly_revenue,
  format(
    lag(revenue) over (order by order_month),
    2,
    'en_US'
  ) as previous_month_revenue,
  format(
    revenue - lag(revenue) over (order by order_month),
    2,
    'en_US'
  ) as month_over_month_change
from monthly_revenue
order by order_month;
```

4. Which product categories generate the most revenue?

```
-- 4. product categories generating the most revenue

select
  p.product_category_name,
  format(sum(oi.price), 2) as category_revenue
from order_items oi
join products p
  on oi.product_id = p.product_id
group by p.product_category_name
order by sum(oi.price) desc;

-- 5. top 3 products within each category by revenue
use olist;
```

5. Rank the top 3 products within each category by revenue.

```
with product_revenue as (select 1 s 836 ms
    p.product_category_name,
    oi.product_id,
    sum(oi.price) as revenue
from order_items oi
join products p
    on oi.product_id = p.product_id
group by
    p.product_category_name,
    oi.product_id),
ranked_products as (select
    product_category_name,
    product_id,
    revenue,
    row_number() over (
        partition by product_category_name
        order by revenue desc) as product_rank
from product_revenue)
select
    product_category_name,
    product_id,
    format(revenue, 2) as product_revenue,
    product_rank
from ranked_products
where product_rank <= 3
order by product_category_name, product_rank;
```

6. Segment customers into spend tiers (Low / Medium / High).

```
-- 6. Segment customers into spend tiers (Low / Medium / High)
#Low: less than 100
#Medium: 100 to below 500
#High: 500 or more

with customer_spend as (select
    c.customer_unique_id,
    sum(op.payment_value) as total_spent
  from customers c
  join orders o
    on c.customer_id = o.customer_id
  join order_payments op
    on o.order_id = op.order_id
  group by c.customer_unique_id)
select
  customer_unique_id,
  format(total_spent, 2) as total_spent,
  case
    when total_spent < 100 then 'Low'
    when total_spent < 500 then 'Medium'
    else 'High'
  end as spend_tier
from customer_spend
order by total_spent desc;
```

7. How many customers are repeat buyers vs one-time buyers?

```
-- 7. how many customers are repeat buyers vs one-time buyers?
with customer_orders as (select
    c.customer_unique_id,
    count(o.order_id) as order_count
  from customers c
  left join orders o
    on c.customer_id = o.customer_id
  group by c.customer_unique_id
  having count(o.order_id) >= 1)
select
  case
    when order_count = 1 then 'One-time buyer'
    else 'Repeat buyer'
  end as buyer_type,
  format(count(*), 0) as number_of_customers
from customer_orders
group by
  case
    when order_count = 1 then 'One-time buyer'
    else 'Repeat buyer'
  end
order by count(*) desc;
```

8. What percent of total revenue comes from the top category?

```
-- 8. what percent of total revenue comes from the top category?
with category_revenue as (
    select
        p.product_category_name,
        sum(oi.price) as revenue
    from order_items oi
    join products p
        on oi.product_id = p.product_id
    group by p.product_category_name
)
select
    product_category_name,
    format(revenue, 2) as category_revenue,
    concat(
        format(revenue / sum(revenue) over () * 100, 2),
        '%'
    ) as percent_of_total_revenue
from category_revenue
order by revenue desc
limit 1;
```